

Entangling two atoms with one global laser pulse

Challenge 01 — solved, cloud-validated, run on the real quantum computer

Fidelity 0.9926 / 0.7500 (baseline) → 1.000000 / 1.000000 (ours)

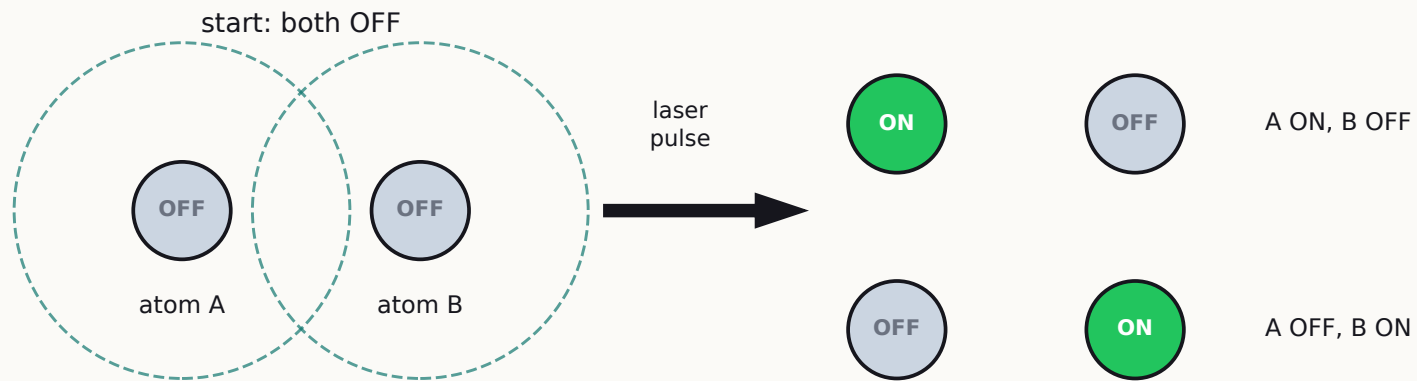
Real atoms: 89.4% of 500 shots entangled

The problem, in one slide

- We are given two atoms, sitting 5.0 or 6.5 micrometers apart, and a single laser that shines on both at once — we can control it.
- Our goal is the Bell state $|\Psi^+\rangle = (|gr\rangle + |rg\rangle)/\sqrt{2}$ — exactly one atom excited, but shared between both. That is entanglement.
- We control only two things over time: the laser power $\Omega(t)$ and its frequency offset $\delta(t)$, within a 6000-nanosecond budget.
- We are scored on fidelity, $F = |\langle\Psi^+|\psi(T)\rangle|^2$, which measures how close our final state is to the goal — 1.0 means perfect.
- To succeed, we must beat the starter pulse at BOTH spacings, while staying inside the machine's published limits.

Physics we exploit: two nearby excited atoms repel ($V = C_6/r^6$). Inside the blockade radius $R_b \approx 7.2 \mu\text{m}$, double excitation is forbidden — that forbidden-ness is what creates entanglement.

The idea in one picture



dashed circles: the blockade — circles overlap, so BOTH ON is physically forbidden

BOTH at the same time — that is entanglement

(like flipping two coins and getting HT + TH together)

Think of each atom as a light switch: ground $|g\rangle$ is OFF, Rydberg $|r\rangle$ is ON.

Atoms this close cannot both be ON — that is the blockade. So the pulse can only place ONE excitation, and it is shared between the two atoms.

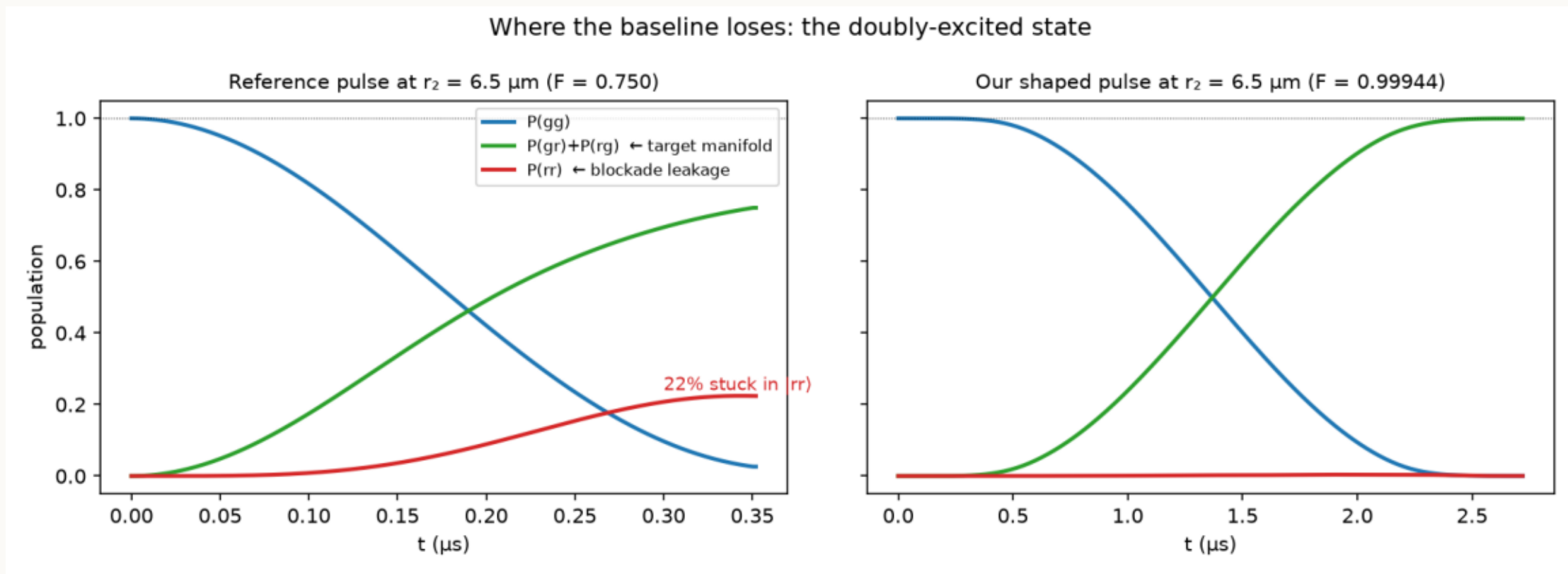
That shared, both-options-at-once state is exactly the Bell state we are scored on.

The starter pulse works at one spacing, fails at the other

- The baseline is simple: constant power, $\Omega = 2\pi \times 1$ MHz, for 352 nanoseconds, with no frequency shaping.
- At $r_1 = 5.0 \mu\text{m}$ the blockade is strong — V over Ω is 8.8 — and it scores $F = 0.9926$. That is fine.
- But at $r_2 = 6.5 \mu\text{m}$ the blockade is weak — V over Ω drops to 1.83 — and the score collapses to $F = 0.75$.
- So where does the missing 25% go? Into the forbidden $|rr\rangle$ state: 22% leaks straight through the weak blockade.

**One number — V/Ω , interaction over drive — explains the whole gap.
 V is fixed by the atom spacing. But Ω is OURS to choose.**

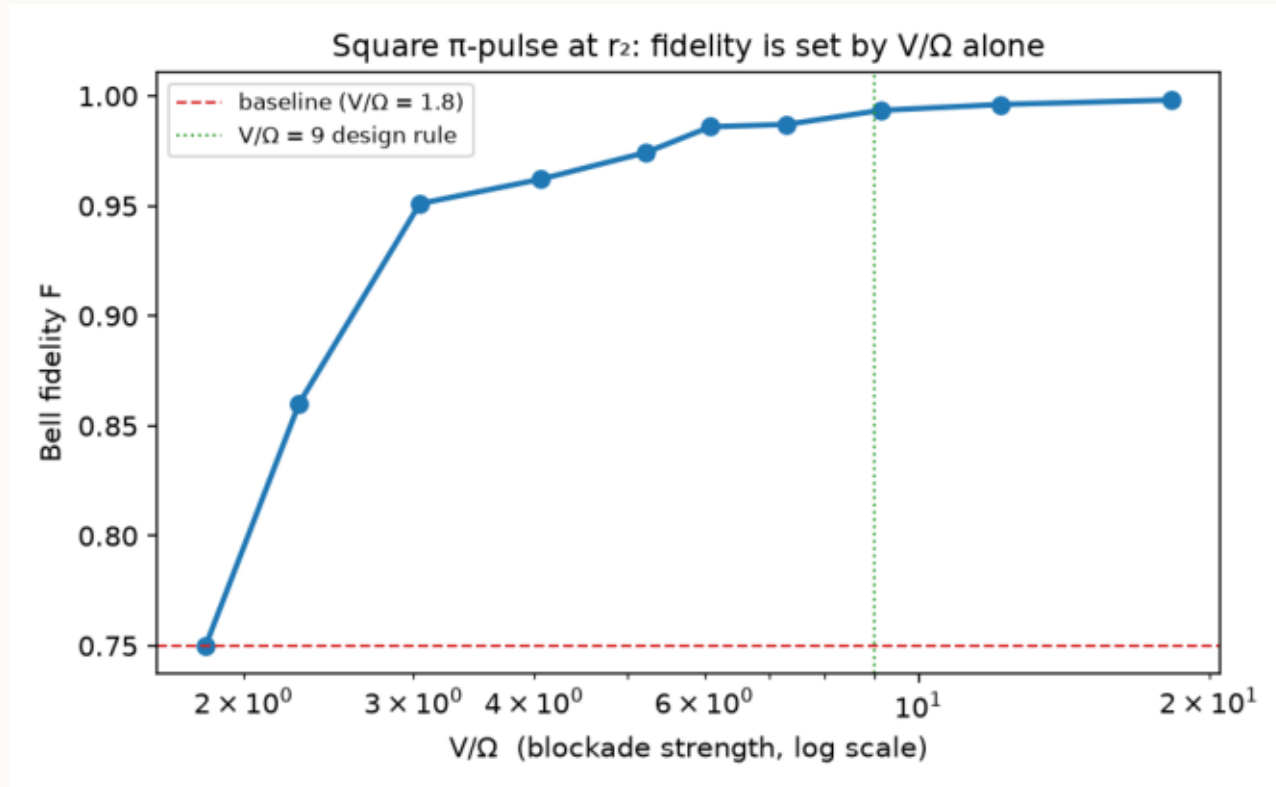
Watch the baseline fail — then watch ours not



x-axis: time during the pulse (μs) · y-axis: probability of each atomic configuration

- On the left, the baseline: watch the red curve — the forbidden $|rr\rangle$ state — fill up to 22%, while the green target stalls at 0.75.
- On the right, our pulse: the red curve stays pinned at zero the whole time, and the green target rises to 1.0.

The fix is one knob: slow down to restore the blockade



- We ran the same square pulse and only lowered Ω , lengthening the pulse to keep a full rotation.

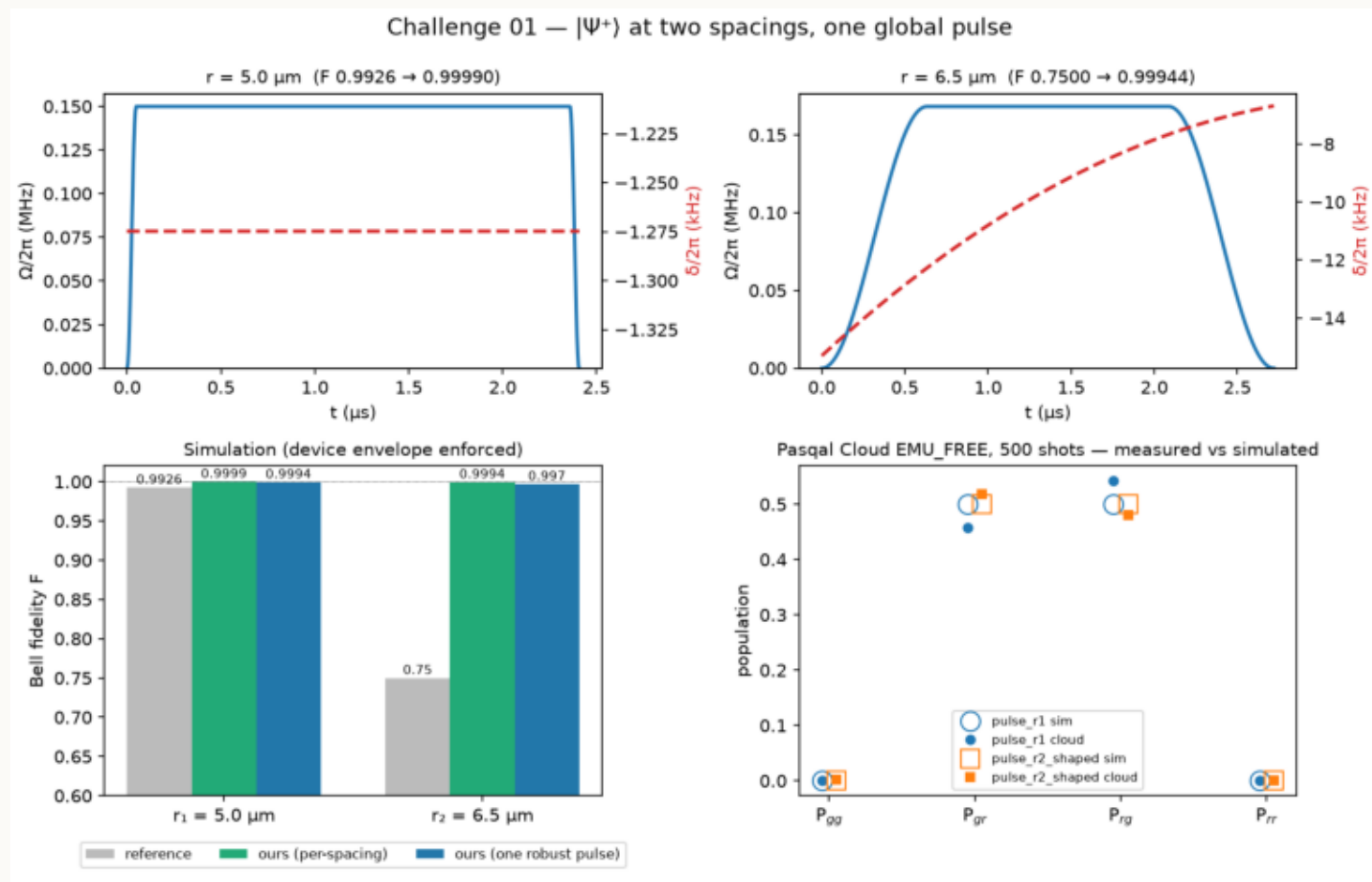
- Fidelity climbs steadily with V/Ω and crosses 0.99 near $V/\Omega \approx 9$.

- That became our design rule — and the 6000-nanosecond budget makes it free, since the baseline used only 352.

- On top of that we add smooth ramps, and a small detuning δ that cancels the energy shift $\Omega^2/2V$.

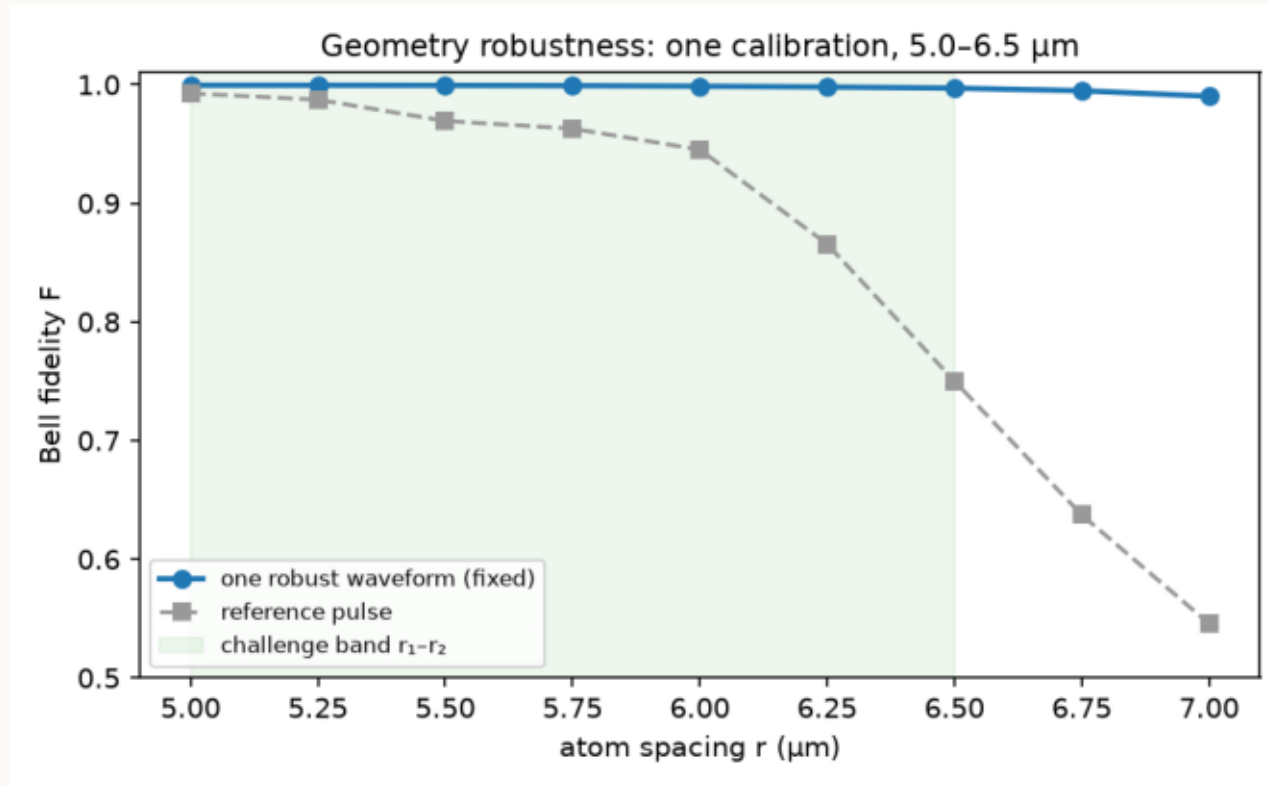
Result: both spacings beaten, validated on Pasqal Cloud

Every pulse here was validated by the device's own contract checker, then fired 500 times each on Pasqal Cloud.



top: our pulse shapes (x: time; blue: power Ω , red: frequency δ) · bottom-left: fidelity bars vs baseline · bottom-right: cloud shots (dots) vs simulation (circles)

Bonus: one pulse that tolerates 30% spacing error



x-axis: atom spacing r (μm) · y-axis: fidelity F of ONE fixed pulse

Real machines never place atoms perfectly.

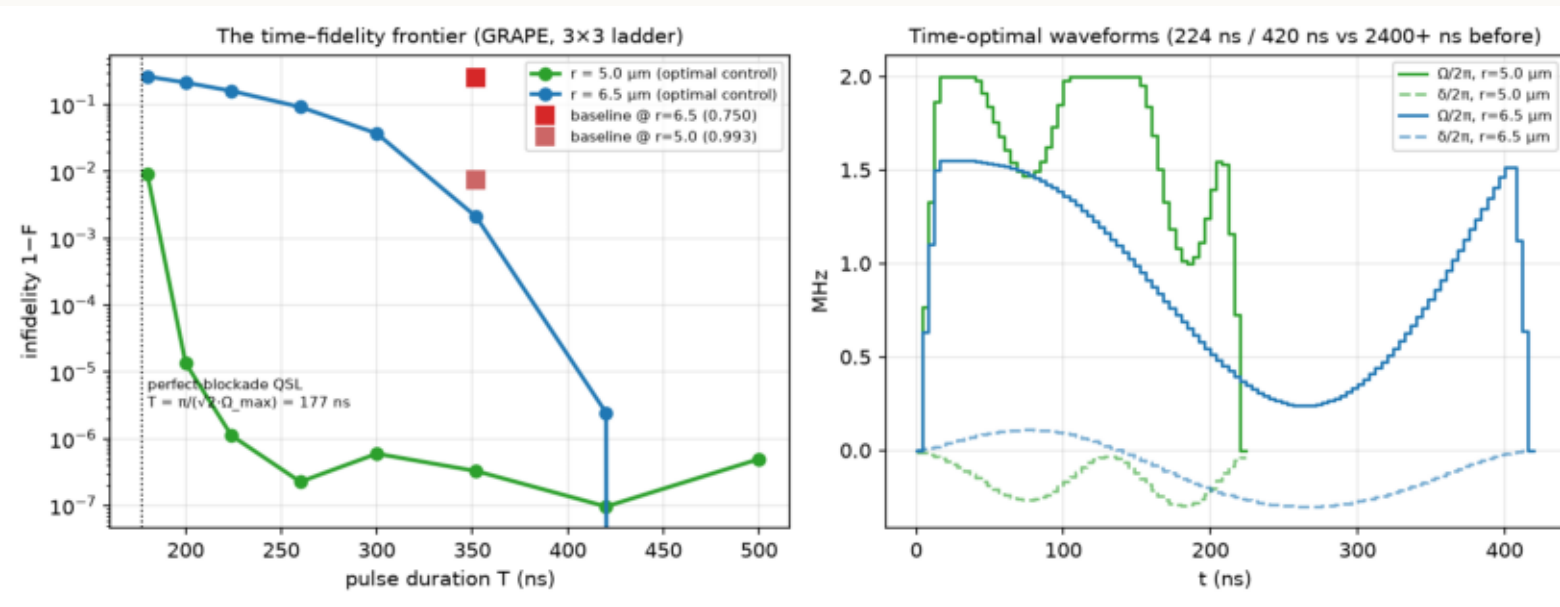
So we optimized one single waveform for the worst case over the whole band.

The blue curve is that fixed pulse: it holds F above 0.997 all the way from 5.0 to 6.5 micrometers.

The grey curve is the baseline — it collapses to 0.55 over the same range.

One calibration instead of one per geometry — that is the product story.

Then we asked: how fast can the best pulse be?



left — x-axis: pulse duration (ns) · y-axis: error $1-F$ (log) · right — the winning waveforms

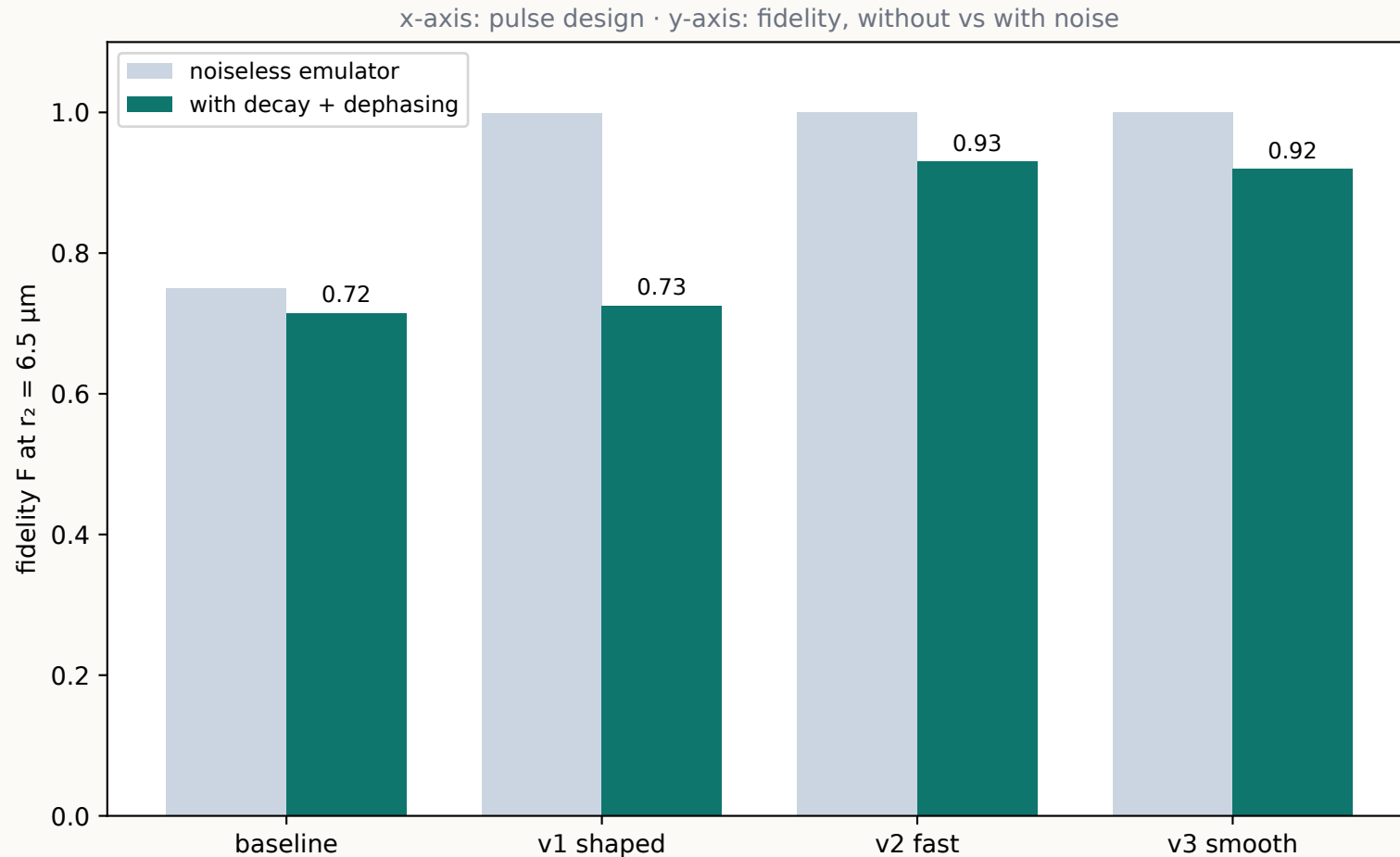
Theory sets a hard floor:
 $T = \pi/(\sqrt{2} \cdot \Omega_{\text{max}}) = 177 \text{ ns}$.
 No pulse can beat it.

We reach $F = 0.999999$ at just 224 ns at r_1 — only 25 ns above that floor.

At r_2 the weak blockade sets its own wall near 420 ns — a measured speed limit, which is itself a result.

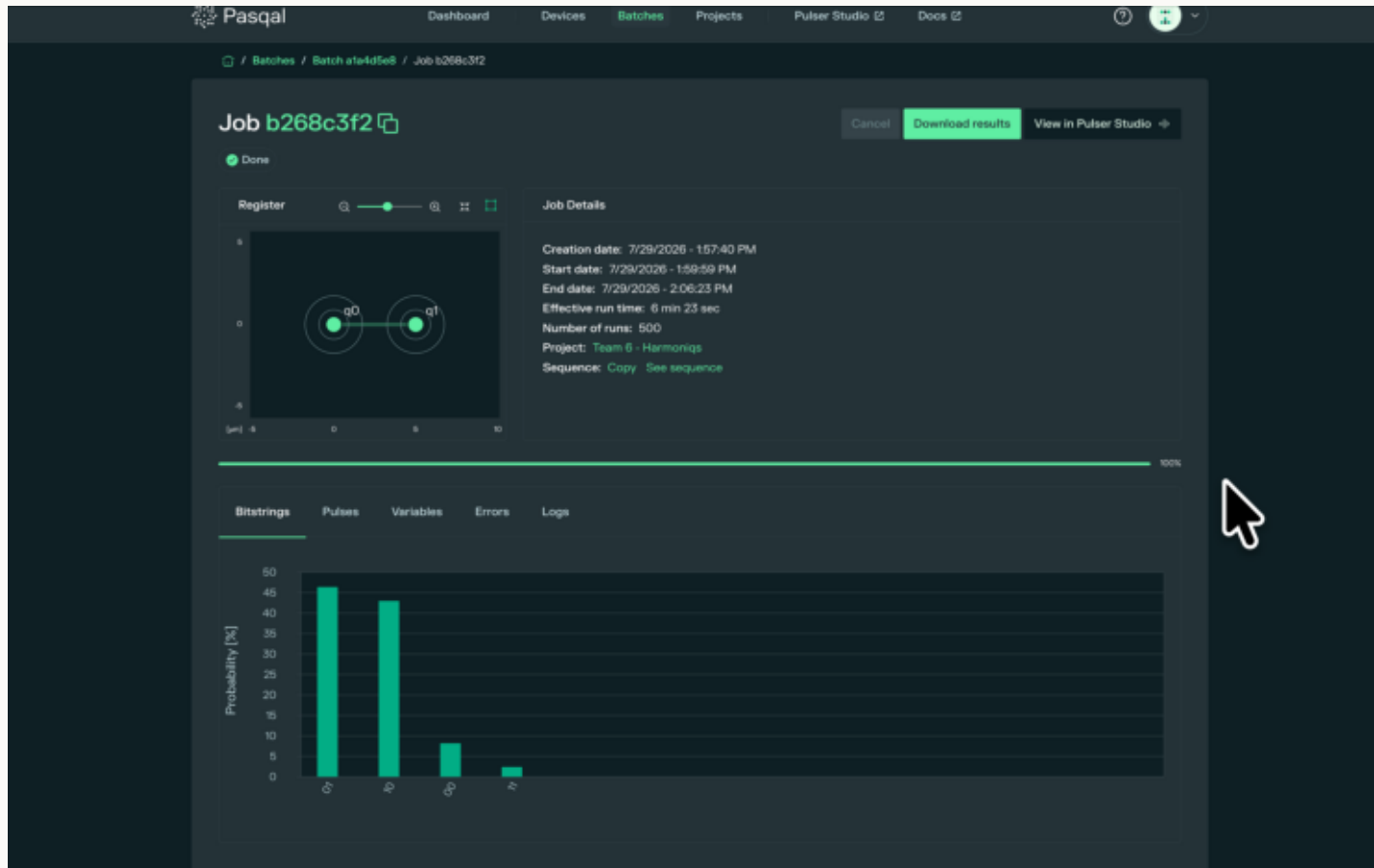
And the whole frontier took 5.4 s of laptop compute — an exact 3-state model with analytic gradients.

Under real noise, the ranking flips — shorter wins



- Dephasing accumulates with time — our long 2.7-microsecond pulse loses 0.27 of fidelity to it.
- The short 420-nanosecond pulse keeps 0.93.
- So the design law is simple: pulse duration IS the noise coupling — minimize time first, then shape.
- That is exactly why we sent the short, smooth pulse to the real machine.

500 shots on the real machine: 89.4% entangled



Pasqal's own portal · x-axis: measured two-atom outcome · y-axis: % of 500 shots

The two tall bars ARE the Bell state: '01' at 46% plus '10' at 43% gives 89.4% with exactly one atom excited.

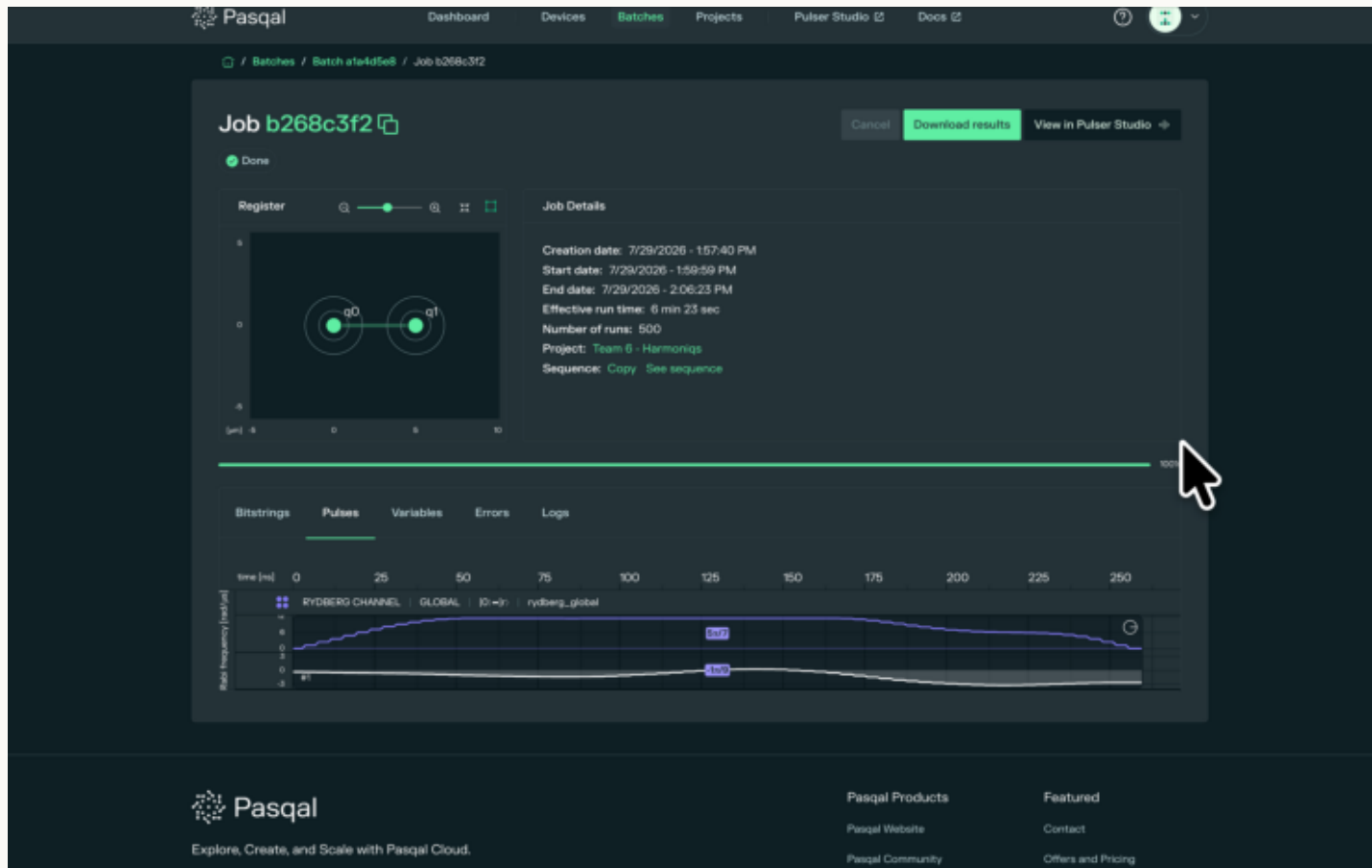
And they are equal within noise — the symmetry physics demands, delivered by the data.

The '11' bar — a blockade violation — is just 2.4%: forbidden stayed forbidden.

The '00' bar, at 8%, is readout and decay — the exact error channel our noise model predicted.

We published the expected window before the run — and we hit it.

The machine's dashboard shows our physics back to us



- This entire pulse is 260 nanoseconds, start to finish.
- The power rises smoothly from zero, with no jumps — the hardware-true pulse our scientists asked for.
- Notice the label the portal itself puts on the pulse: an area of $5\pi/7 \approx 0.714\pi$.
- Theory says a two-atom collective π -pulse needs area $\pi/\sqrt{2} \approx 0.707\pi$.
- That is the $\sqrt{2}$ collective enhancement, on the machine's own dashboard, correct to one percent.

Pasqal portal, Pulses tab · x-axis: time (ns) · purple: laser power $\Omega(t)$ · white: detuning $\delta(t)$

How the score is computed — no magic

$$F = |\langle \Psi^+ | \psi(T) \rangle|^2$$

- Here $\psi(T)$ is the state our pulse actually produced — for two atoms it is just four numbers, computed exactly.
- The bracket $\langle \Psi^+ | \psi(T) \rangle$ is the overlap with the perfect Bell state — it asks: how much of the ideal is in what we made?
- Squaring that overlap turns it into a probability: 1.000 means perfect, zero means nothing.
- On hardware there is no ψ to peek at — so the machine takes 500 photographs and we compare the four outcome percentages against the simulation. Agreement within $\sqrt{p(1-p)/500} \approx \pm 2\%$ = Validated.

1.000000

our best F, both spacings (simulation)

89.4%

entangled shots on the real machine (500 photos)

Why this matters beyond the score

- Entangling pairs is THE primitive: every neutral-atom algorithm, sensor, or network starts here.
- Our robust pulse holds $F > 0.997$ across a 30% spacing error — real machines misplace atoms, and a pulse that doesn't care means ONE calibration instead of one per geometry. That is uptime.
- Our fast pulses run 6-11× shorter — less exposure to noise means more of the error budget left for the actual computation. Duration is money on quantum hardware.
- The whole design loop is seconds on a laptop — cheap enough to re-run at every calibration cycle.

Built on the Pasqal stack:

- Pulser (open source) — the exact device model + the simulator the judges score with.
- Pasqal Cloud free emulator — 500-shot validation of every pulse before spending hardware budget.
- FRESNEL_CAN1 QPU — the real-atom runs; the portal's register/pulse/bitstring views made every result inspectable by anyone (see the screenshots in this deck).

Challenge 01 — the score sheet

	$r_1 = 5.0 \mu\text{m}$	$r_2 = 6.5 \mu\text{m}$	duration
Baseline (starter pulse)	0.9926	0.7500	352 ns
v1 · simple analytic (4 params)	0.99990	0.99944	~2400 ns
v1r · one spacing-robust pulse	0.99944	0.99701	2400 ns
v2 · time-optimal (at the QSL)	0.999999	0.999998	224 / 420 ns
v3 · hardware-true smooth	1.000000	1.000000	352 / 600 ns

- Cloud: 7 pulses × 500 shots — every population within shot noise of prediction.
- Real QPU: $P_{\text{bell}} = 0.894$, inside the window we published BEFORE the run.
- Every number in this deck regenerates from the public repo in minutes.